

LEADING FROM THE FRONT PM Narendra Modi meeting the injured passengers on June 3, (right) railway minister Ashwini Vaishnaw offers prayers as services resume in Balasore, June 4

software-driven, EIS clears routes for approaching trains, saving time as well as ensuring safety of movement by eliminating almost all manual relays and interconnections. The system has three main components: the switching points to direct a train to a loop line if needed, a track occupancy sensor, and signals to control train movement and coordinate traffic.

Talking to key railways personnel, INDIA TODAY has pieced together the sequence of events that took place on that fatal day at Bahanaga Bazar. Here is what seems to have transpired:

4.35 pm: The signal inspector of the section, whose job it is to oversee signal maintenance, approaches the station master with a hand-written memo to finish replacing the manual boom barrier at the railway crossing nearby with the more lightweight Electric Lifting Barrier (ELB). Button-operated, ELBs are far more efficient than their mechanically operated counterparts, and are part of the railways upgradation project. The station master authorises what is called a 'disconnect', which delinks the boom barrier movement from the signal interlocking system. This disconnect permits trains to keep moving on the main line even while repair work is going on for the boom barrier.

6.35 pm: An iron ore-laden goods train travelling on the Up line in the Chennai direction is directed to the loop line so that the superfast Coromandel Express, which is due 20 minutes later, can use the main line unhindered.

Earlier, the station master at the Bahanaga Bazar station has called the boom barrier guard in the hut on the dedicated phone line to tell him not to allow any vehicular movement at the crossing till asked to do so. Meanwhile, the signal inspector, assisted by a couple of electrical signal maintainers and gang workers, is giving the final touches to the installation of the barrier. They had mounted the concrete pedestal to support the barrier a few days before and laid fresh cables connecting it to the location box nearby that contains the row of relay panels linking it to the EIS.

6.45 pm: The signal maintainers replace the old black-coloured wires in the six relay switches with bright red ones and test the barrier system. It is at this point, it is believed, that the signal system is 'interfered with', which subsequently causes errors in the signalling commands.

6.51 pm: The signal inspector then informs the station master that they have finished their work and that the boom barrier has been reconnected to the main signal interlocking system. But the station master fails to do the mandatory physical verification to ensure that the signal is working normally. This proves to be a fatal mistake. Instead, the station master just pushes the button that activates the Distant Signal, located 2 km away from the station, to turn green to allow the Coromandel Express travelling on the main line to pass through.

A similar incident of errant signalling had **occurred in South Western Railway zone in February**, but no action was taken

6.53 pm: The loco pilot of the Coromandel Express sees the green signal and continues at 128 kmph, just two kmph short of the 130 kmph permitted while entering the station jurisdiction. The Home Signal, just 180 metres before the loop line at the station, is also showing green, allowing the train to proceed at the same speed. The loco pilot has no indication at this point that he will soon be diverted to the loop line. Inexplicably, the interlocking system activates the loop line provision at Point 17A of the switch board. This, in turn, commands the point blades—a pair of tapering rails lying between the diverging outer rails—to move and direct the train towards the loop line.

6.55 pm: Within seconds, the Coromandel Express slams into the rear wagons of the stationary goods train, the collision at 128 kmph pushing the engine almost vertically upwards and sending six of the 23 coaches cartwheeling in the air. These, in turn, catch two coaches at the tail end of the Yeshwantpur-Howrah Express travelling in the opposite direction and running three hours late, causing that train to derail as well.

The site of the railway accident looked as though "a nuclear blast" had ripped through the landscape, as one bystander put it. The two trains had been ferrying 2,296 passengers and nearly 1,500 were hit—such was the magnitude of the impact. As multiple agencies mounted rescue and relief measures, the Commission for Railway Safety (CRS), an autonomous body